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Supine leg press as an alternative to standing lunge in high-speed stereo radiography

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ABSTRACT

The standing lunge is an activity commonly used to quantify in-vivo knee kinematics with fluoroscopy. The ability to perform the standing lunge varies between subjects and can necessitate movement accommodations to successfully complete the desired range of motion. We proposed a supine leg press as an alternative to the standing lunge that aimed to provide a similar evaluation of knee motion while increasing the measured range of motion. Tibiofemoral kinematics of 53 non-symptomatic adults (27 men, 26 women, 50.8 ± 7.0 yrs.) were calculated from the tracked high-speed stereo radiography (HSSR) images for supine leg press and standing lunge using CT-segmented bony geometries of the right lower limb. The supine leg press proved to be a useful alternative to the standing lunge while providing 46.2° greater range of motion in knee flexion. The difference in angle-matched kinematics across a 100° flexion range between the leg press and lunge was 0.70° in varus-valgus rotation, 1.5° in internal-external rotation, 1.0 mm in medial–lateral translation, 2.3 mm in anterior-posterior translation, and 0.46 mm in superior-inferior translation for men. The angle-matched difference for women across 100° was 0.58° in varus-valgus rotation, 2.4° internal-external rotation, 0.70 mm medial–lateral translation, 2.1 mm anterior-posterior translation, and 0.78 mm superior-inferior translation. The similar kinematics, while having a greater range of motion, and control of the applied load makes the supine leg press an alternative for quantifying in-vivo knee kinematics.

1. Introduction

The use of fluoroscopy to evaluate the impact of knee pathology and treatment on joint kinematics has become widespread. Most laboratories rely on patients performing a standing lunge for measurement of loaded knee kinematics while maintaining the knee in the field of view of the fluoroscopy system. This weightbearing activity has been formative in basic knee research, evaluation of knee treatment, and knee arthroplasty design (Cooke et al., 1997; Defrate et al., 2004; Hamai et al., 2013, 2009; Kefala et al., 2021; Leszko et al., 2011; Navacchia et al., 2017). However, instruction of the lunge varies between research settings with some collecting data at discrete knee angles (Defrate et al., 2004; Qi et al., 2013), continuously through a functional range of motion (Feng et al., 2015; Komistek et al., 2003; Moro-Oka et al., 2008; Navacchia et al., 2017), or adding a forward lean to facilitate observations at deep-knee angles (Hamai et al., 2013, 2009; Leszko et al., 2011). Moreover, some participants may require balance support or direct assistance to ensure the knee remains in the fluoroscopic field of view during lunge, which may confound their full voluntary range of knee-specific motion.

Despite these challenges, a 2017 *meta*-analysis of 12 studies including 164 non-symptomatic participants aged 25 to 61 demonstrated that squatting, lunging, and kneeling provided similar kinematic profiles between 120° and 135° (Galvin et al., 2018); but this range of motion may not be obtainable for all populations. For example, Kefala et al. (2021) measured the knee kinematics of subjects with total knee replacement during single-leg lunge up to 80°, as many subjects could not perform deeper flexion angles during the activity. Common activities of daily living such as rising from a chair and using stairs require a knee that flexes 90°-120° (Rowe et al., 2000) while more demanding activities, including using the bath (135°; Rowe et al., 2000) and getting up from the floor, necessitate up to 140° of knee flexion (Weiss et al., 2002).

Enabling the evaluation of weightbearing knee function,

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independent of confounding patient factors, in deeper knee flexion is critical because range of motion is a strong predictor of post-operative functional ability (Bade et al., 2014; Ha et al., 2016) and increases in range of motion are positively associated with patient satisfaction (Devers et al., 2011; Ha et al., 2016). Deeper flexion measures are not exclusively useful to osteoarthritic and total-knee arthroplasty populations but are valuable in studying anterior cruciate ligament reconstruction (Shelbourne et al., 2012), cerebral palsy gait deviations (Papageorgiou et al., 2019), and general knee pathologies (Zhang et al., 2020). Furthermore, understanding how individual limitations and variability contribute significantly to knee kinematics, markedly in deep flexion, is vital for patient-specific approaches to kinematic computational modeling (Fitzpatrick et al., 2012).

Measurement of knee kinematics with a leg press seeks to isolate functional knee movement with a repeatable and accessible activity. The supine leg press allows for control of the load supported by the foot (Escamilla et al., 2001; Fleming et al., 2005; Scarvell et al., 2004), does not require the contralateral limb to perform the task (Macaluso and De Vito, 2003), and does not require the assistive use of hands. Instead, only the limb of interest is in contact with the foot plate and is solely used to generate the force to complete the activity. Due to the challenges of deep-knee flexion under weightbearing conditions, we propose a supine leg press as an alternative to the lunge that aims to provide a similar evaluation of knee motion while increasing the measured range of motion. The purpose of this study was to verify the use of a supine leg press as an alternative to the standing lunge in a non-symptomatic adult cohort and identify potential differences between these tasks.

2. Materials and methods

Fifty-four non-symptomatic participants (27 women, 50.8 ± 7.0 yrs., 1.75 ± 0.1 m, 79.1 ± 15.4 kg) underwent a unilateral-lower extremity CT scan and 1-hour protocol that consisted of 6 tasks in high-speed stereo radiography (HSSR) for the right knee. These tasks included supine leg press (20 lbs. (9.1 kg); Fig. 1A), seated-knee extension, single-leg stance, double-leg stance, standing lunge (Fig. 1B), and gait. A combined effective dose of 1.01 mSv (0.16 mSv from knee CT (Biswas et al., 2009), 0.75 mSv from the lower limb scout CT (Henckel et al., 2006), and 0.10 mSv from stereo fluoroscopy) was estimated for each participant. For the purposes of this manuscript, only the supine leg press and standing lunge were analyzed. All participants provided informed consent to a protocol approved by the Institutional Review Board of the University of Denver (IRBNet ID: 1441095).

A HSSR imaging system (Ivester et al., 2015) was used to quantify tibiofemoral knee kinematics. The system was comprised of two 40-cm image intensifiers integrated with high-speed, high-definition digital cameras placed at an approximate 70° relative angle. Activities were captured at 50 frames/s using pulsed radiography for each plane. Marker-based motion-capture techniques were collected (100 Hz sampling frequency; VICON, Centennial, CO) and forces measured from an indwelling force plate (1000 Hz sampling rate; Bertec, Columbus, OH). All measures were recorded concurrently and managed by Vicon Nexus (version 2.9.3).

Due to the limited HSSR capture volume, the leg press was completed in two trials to record the full range of motion about the knee flexion axis: one to record deep flexion and a second to record terminal knee extension. The leg-press apparatus featured a rolling sled attached to a hanging 20-lb weight (9.1 kg) by a cable and series of pulleys (Fig. 1A). The 20-lb weight was selected to provide sufficient resistance while enabling easy completion by all participants. Placement of the apparatus on a hydraulic lift table allowed the participants to remain in place as the height of the table was raised and the trial repeated with as little disturbance as possible. Participants were instructed to lie supine on the sled and place their right foot on an elevated platform in a neutral, toes-up position. Participants began in deep-knee flexion (Fig. 1A *top*) and pushed with their heel to full extension (Fig. 1A *bottom*) with the recording lasting ~ 3 s. Between trials the participants were given verbal instruction to keep their foot in the same location as height adjustments were made and the activity repeated. These separate trials were later combined to be analyzed as one continuous motion during post processing, described later.

The standing lunge was completed in one trial where participants began in a split stance with their right knee forward in the center of the HSSR capture volume (Fig. 1B *top*). They were instructed to lower their hips across 3 s until their back knee touched the ground (Fig. 1B *bottom*). Emphasis was placed on maintaining a continuous view of the right knee for the duration of the HSSR recording while moving across the largest flexion range possible.

Subject-specific bone geometries were segmented from the lower-extremity CT scan consisting of the right femur, a combined tibia and fibula, and patella (Fig. 1C; Simpleware ScanIP, version 02018.12). Bony landmarks were identified for each bone by the same researcher and used to assign the local coordinate system. The tibia and patella local-coordinate systems were assigned according to Grood and Suntay (1983), whereas the femoral-coordinate system assignment was performed using a modified method similar to Tashman et al. (2004).

The femoral-coordinate system identified the femoral-head center with a fitted sphere and used this superior point in creating the long, superior axis. The anterior-posterior axis of the femur was temporarily drawn perpendicular to the long axis and transepicondylar axis (Churchill et al., 1998). The femur geometry was then rotated about the transepicondylar axis to identify the most distal points on the medial and lateral condylar surfaces with circles fit to these points parallel to the sagittal plane. The circle centers were used to assign the medial–lateral (x) axis with the origin set to the mid-point between the two circle centers (Fig. 2). The anterior-posterior (y) axis was drawn perpendicular

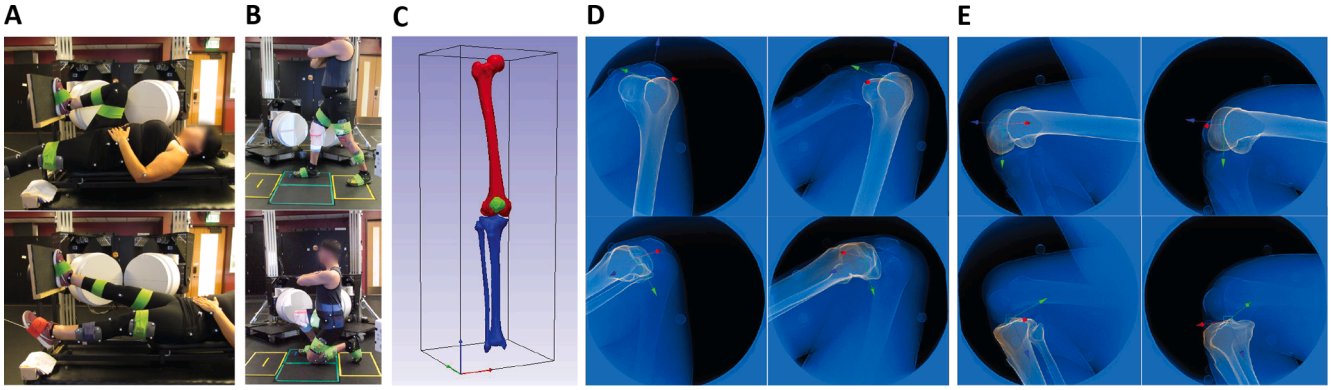

Fig. 1. (A) Supine leg press (B) Standing lunge (C) Right-leg segmentation (D) Stereo tracking of leg press (femur top, tibia bottom) (E) Stereo tracking of lunge (femur top, tibia bottom).

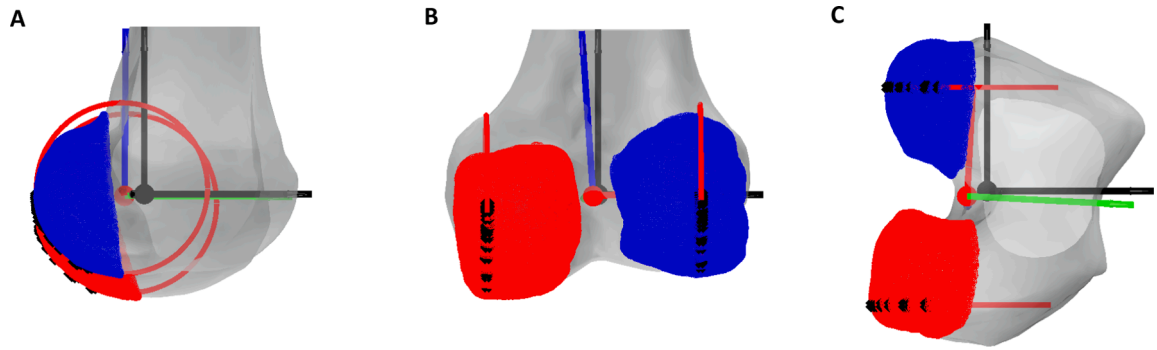

Fig. 2. Femoral coordinate assignment. Medial (red) and lateral (blue) condyles fit with circles used to approximate the optimal flexion axis of the femur. Trans-epicondyal axis (TEA) alignment in black and final alignment in red. (A) Sagittal view; (B) Posterior view; (C) Bottom view. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to the long and medial–lateral axis with the superior-inferior (z) axis drawn perpendicular to the anterior-posterior and medial–lateral axes.

Bone geometries were manually tracked in the two HSSR imaging planes to recreate subject-specific movement in 3D space for each task (Fig. 1D; Autoscoper, Brown University). When all three bones were present in each imaging plane, every third frame of the HSSR recording was tracked resulting in 41 ± 12 [38, 44] manually tracked frames during deep flexion, 23 ± 6 [22, 25] frames during terminal extension,

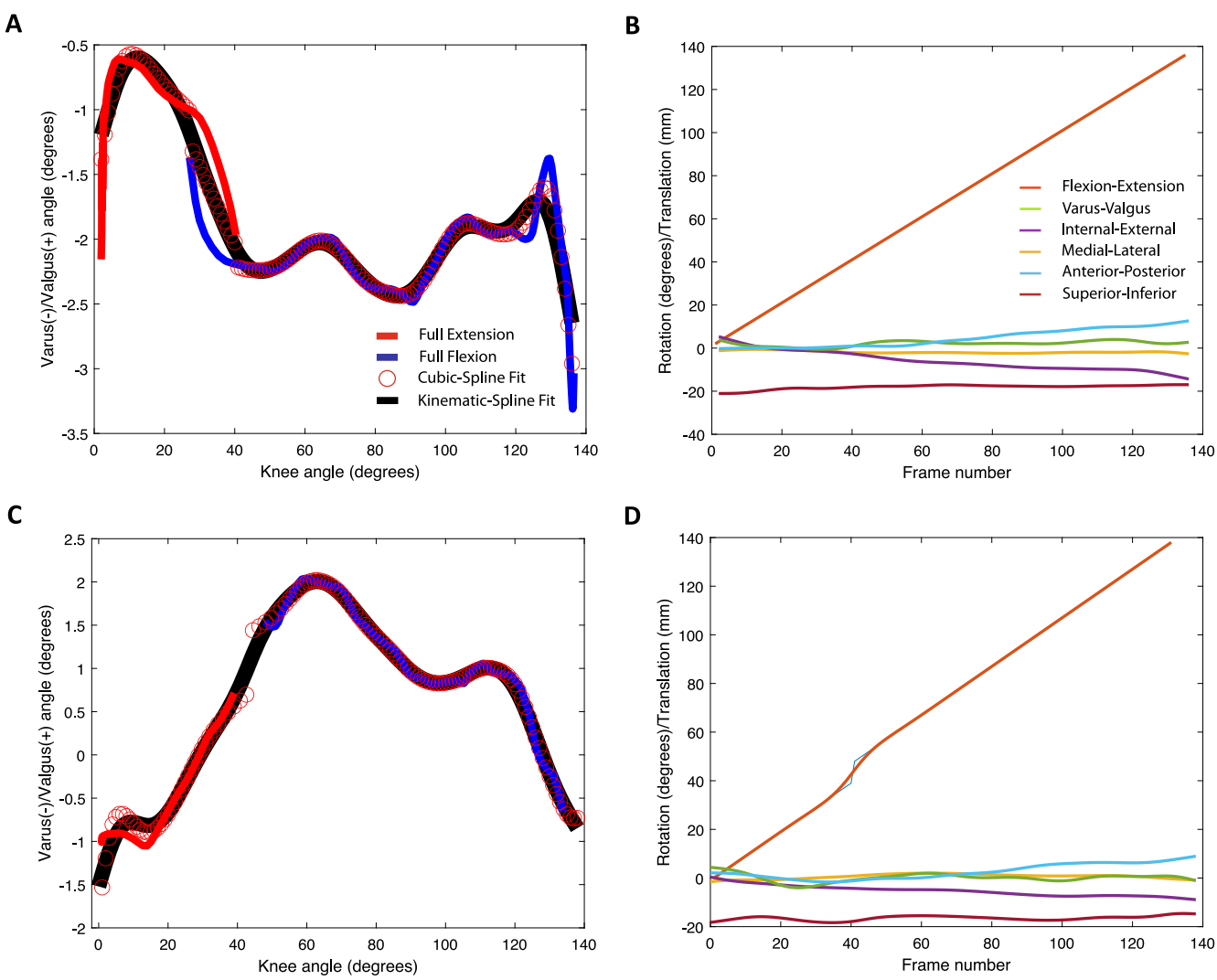

Fig. 3. Leg-press spline fit. Representative data for two participants (A and B, and C and D, respectively) showing the process to generate continuous kinematics from two separate leg-press trials. The red line represents the full extension portion of the leg press and the blue line represents the fully flexed portion. The red circles represent the cubic-spline fit for every 1° and the black line represents the final kinematic spline. (A) Varus/Valgus spline for overlapping trials; (B) Six degrees of freedom corresponding with spline of A; (C) Varus/Valgus spline for non-overlapping trials; (D) Six degrees of freedom corresponding with spline of C. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and 44 ± 22 [38, 50] frames during lunge. Bone position was interpolated between manually tracked frames using the built-in algorithm in Autoscoper. Interpolation resulted in 124 ± 36 [114, 133] tracked frames during deep flexion, 69 ± 18 [65, 74] tracked frames during terminal extension, and 133 ± 65 [115, 151] tracked frames during lunge. Kinematics were calculated about the knee for six degrees of freedom using the Grood and Suntay (1983) convention. A third-order one-dimensional median filter (*y = medfilt1(x)*, MATLAB, version R2019b) was applied to all kinematics to remove rotations and translations outside the physiological range. The median filtered tibiofemoral kinematics of all trials were resampled with cubic splines to provide results in 1° increments across the full knee-flexion range of motion. This procedure resulted in an average 87 ± 22 [81, 93] data points during lunge with additional processing performed to combine separate leg-press kinematic trials (Fig. 3).

The combination of the two leg-press trials into one tibiofemoral kinematic profile consisted of quantifying overlap and smoothing the transition between trials. In the instances when flexion angles overlapped between both trials, the average of the six degrees-of-freedom kinematics was taken at each common flexion angle (Fig. 3B; n = 35; overlap = 18.0 ± 12.9° [13.6, 22.5]; max = 50.2°, min. = 0.6°). In the instances when overlap of flexion was not included, a cubic spline interpolation was applied (Fig. 3C and 3D; n = 18, flexion gap = 11.7 ± 5.2° [9.1, 14.3]; max = 20.1°, min. = 2.0°). In total, each trial was median filtered, resampled to 1° knee flexion increments, and the transition between trials was resampled and discontinuities smoothed. The subsequent filtering and resampling resulted in an average 133 ± 9 [130, 135] data points for the continuous leg press. Only the continuous, resampled leg-press and lunge kinematics were used for analysis.

To assess the kinematic differences in an alternate way, lunge kinematics were subtracted from the leg press at identical knee-flexion angles through each subject's angle-matched flexion range. Only matching flexion angles between the lunge and leg press were used, truncating flexion angles not present in both activities. The angle-matched kinematics were transformed to tracked data in 3D space with subject-specific femurs aligned so both activities would have the same reference femur. The displacement of the tibia during lunge relative to the tibia during leg press was calculated at each common flexion angle. The tibial displacement relative to the common reference femur was quantified using the tibiofemoral Grood and Suntay (1983) kinematic convention to provide angle-matched rotational and translational displacements between the two activities.

Normality of data was assessed using the Shapiro-Wilk test and verified visually with box plots and quantile–quantile plots (theoretical vs. sample quantiles). Descriptive statistics of men and women were compared using parametric t tests and verified with linear regression. Kinematic comparisons between activities were analyzed using linear mixed-effects analysis controlling for random (subject) and fixed effects to control for individual variability between activities. Due to the significant differences and interactions between men and women, sex-specific models were used to quantify the differences in kinematics between the supine leg press and standing lunge. A forward selection procedure was used to evaluate significance of orthogonal-polynomial-regression coefficients for each activity, degree of freedom, and address collinearity of polynomial coefficients. Final models required each order of the hierarchical polynomial model (1st – 3rd) to be significant. The model with the lowest Aikaike Information Criteria was selected with the residuals plotted and visually inspected to determine appropriate fit. Knee kinematics were described for the supine leg press and standing lunge using linear regression (varus-valgus angle), 2nd-order polynomial regression (internal-external angle, medial–lateral translation, and anterior-posterior translation), and 3rd-order polynomial regression (superior-inferior translation). Raw-polynomial regression results were reported to facilitate coefficient interpretation. The effect of knee angle on the displacement of the subtracted kinematics was quantified using simple linear regression. All statistical procedures were performed using R (version 3.6.1) with α set at 0.05.

3. Results

Results are presented for 53 participants (n = 54 total) (Table 1) with one subject removed from analysis due to their inability to complete all activities and suspected osteoarthritis upon review of their CT scan. The analyzed men were 4 years older than the women, 0.15 m taller, and 16.1 kg heavier with no difference in body mass index (BMI; Table 1).

The average range of motion of knee-flexion angle was 133.5 ± 7.6° (range: 107°-147°) during the supine leg press with no significant difference between men and women (Table 2). Men were 4.16° more varus than women at 0° knee flexion with a similar valgus rotation with increased knee flexion between sexes (supplemental materials eq. S1). Additionally, men were 3.19° more externally rotated relative to women at 0° knee flexion (eq. S2b) with sex-specific internal tibial rotation associated with increased knee flexion (predicted values at 100° women = -12.7° [-12.9, 12.5]; men = -14.3° [-14.5, −14.1]). No significant difference between sexes was identified for medial–lateral (eq. S3b) or anterior-posterior translation (eq. S4b) across the flexion range. The tibias of men were 0.66 mm closer to the femur (superior) compared with women, while controlling for height (-11.64 mm [-12.42, −10.83], $P < 0.001$), in superior-inferior translation (eq. S5b).

The average range of motion for knee-flexion angle was 87.1° ± 22.4° during standing lunge with no significant difference between men and women (Table 2). At 0° knee flexion, the tibias of men relative to women were 4.42° more varus (eq. S6), 2.65° externally rotated (eq. S7b), 0.96 mm more medial (eq. S8b), 1.9 mm more posterior (eq. S9b), and 1.47 mm inferior from the femur while controlling for height (-12.16 mm [-12.48, −9.51], $P < 0.001$; eq. S10b). Increased knee flexion during lunge was associated with sex-specific valgus rotation (predicted values at 100° women = 1.15° [0.96, 1.33]; men = -2.20° [-2.38, −2.02]; eq. S6), similar internal rotation (eq. S7b), sex-specific medial translation (predicted values at 100° women = 0.37 mm [0.26, 0.48]; men = -0.21 mm [-0.31, −0.10]; eq. S8b), sex-specific anterior translation (predicted values at 100° women = 8.61 mm [8.42, 8.79]; men = 8.61 mm [8.43, 8.79]; eq. S9b), and similar superior translation of the tibia (eq. S10b).

Men had a 49.5° ([39.9, 59.2], $P < 0.001$) greater knee-flexion range during leg press than lunge. Additionally, the tibias of men relative to their respective femurs were 1.48° more varus (eq. S11) and 0.84 mm more medial during lunge at 0° knee flexion (Fig. 4A). The effect of knee angle varied between activities, and across degrees of freedom. Every degree increase in knee angle during the lunge, compared with the leg press, was associated with an additional 0.019° valgus rotation (eq. S11), 0.048 mm less medial translation and 0.00039° less convex activity-by-knee angle relation (eq. S13b), 0.036 mm additional anterior translation and 0.00029° less convex activity-by-knee angle relation (eq. S14b), and a greater curvilinear activity-by-knee angle relation in superior-inferior translation (eq. S15b).

Women had a 45.3° ([37.8, 52.7], $P < 0.001$) greater knee-flexion range during leg press than lunge. The tibia relative to the femur at 0° knee flexion during lunge was 1.19° more varus (eq. S16), 2.73° internally rotated (eq. S17b), 0.40 mm laterally translated (eq. S18b),

Table 1
Descriptive statistics.

	Men (n = 27)	Women (n = 26)	Difference	P value
Age (yrs.)	53 ± 8	49 ± 5	4 [0.1, 7.6]	0.041
Height (m)	1.82 ± 0.05	1.67 ± 0.07	0.15 [0.12, 0.19]	1.07E-11
Weight (kg)	87.0 ± 15.0	70.9 ± 11.2	16.1 [8.8, 23.3]	5.07E-05
BMI	26.2 ± 3.02	25.5 ± 5.4	0.7 [-1.7, 3.1]	0.575

Mean ± standard deviation [95% confidence interval]; BMI = body-mass index.

Table 2
Range of motion.

Supine-Leg Press	Men	Women	Difference	P value
Flexion-Extension (°)	134.7 ± 5.9	132.3 ± 9.0	2.39 [-1.78, 6.56]	0.254
Varus-Valgus (°)	6.0 ± 2.5	6.2 ± 2.5	−0.21 [-1.59, 1.17]	0.762
Internal-External (°)	18.5 ± 7.0	14.9 ± 4.1	3.63 [0.44, 6.82]	0.0264
Medial-Lateral (mm)	6.7 ± 1.4	5.9 ± 1.7	0.79 [-0.07, 1.66]	0.0714
Anterior-Posterior (mm)	12.3 ± 4.5	10.8 ± 3.6	1.52 [-0.72, 3.76]	0.179
Superior-Inferior (mm)	5.9 ± 1.8	5.3 ± 1.6	0.59 [-0.34, 1.52]	0.207
Standing Lunge	**Men**	**Women**	**Difference**	**P value**
Flexion-Extension (°)	86.3 ± 24.6	88.2 ± 20.6	−1.85 [-14.38, 10.69]	0.769
Varus-Valgus (°)	4.9 ± 2.0	5.0 ± 1.9	−0.093 [-1.15, 0.97]	0.861
Internal-External (°)	11.6 ± 7.0	9.4 ± 3.8	2.18 [-0.97, 5.32]	0.17
Medial-Lateral (mm)	4.9 ± 1.5	4.7 ± 1.5	0.16 [-0.68, 0.99]	0.709
Anterior-Posterior (mm)	8.9 ± 4.6	7.6 ± 2.4	1.34 [-0.71, 3.40]	0.195
Superior-Inferior (mm)	4.5 ± 1.6	4.4 ± 1.8	0.14 [-0.80, 1.08]	0.771

Mean ± standard deviation [95% confidence interval].

2.63 mm anteriorly translated (eq. S19b), and similar in superior-inferior translation compared with the leg press (Fig. 4B; eq. S20b). Similar to men, the effect of knee angle for women varied between activities and across degrees of freedom. Every degree increase in knee angle when comparing the lunge to leg press was associated with an additional 0.0082° valgus rotation (eq. S16), a 0.080° less internal rotation and 0.00061° less convex activity-by-knee angle relation (eq. S17b), a 0.005 mm less anterior translation (eq. S19b), and a greater curvilinear activity-by-knee angle relation in superior-inferior translation (eq. S20b).

The subject-specific subtractions in tibiofemoral rotations and translations between the leg press and lunge were quite small across the functional range. The angle-matched displacements for men showed that for each degree increase in knee-flexion angle there was a 0.0070° ([0.0037, 0.0103], $P = 3.31E\text{-}5$) displacement in valgus rotation, 0.015° ([-0.021, −0.010], $P = 4.92E\text{-}8$) internal rotation, 0.010 mm ([-0.014, −0.007], $P = 7.99E\text{-}10$) medial translation, 0.023 mm ([-0.026, −0.021], $P < 2E\text{-}16$) posterior translation, and 0.0046 mm ([0.0030, 0.0062], $P = 2.85E\text{-}8$) superior translation difference (Fig. 5A). These results highlight that across a 100° range of knee flexion the physical displacement of the tibia relative to the femur would be 0.70° in varus-valgus rotation, 1.5° in internal-external rotation, 1.0 mm in medial–-lateral translation, 2.3 mm in anterior-posterior translation, and 0.46 mm in superior-inferior translation.

The angle-matched subtractions for women found with every degree increase in knee-flexion angle there was a 0.0058° ([0.0023, 0.0093], $P = 0.00105$) displacement in valgus rotation, 0.024° ([-0.029, −0.018], $P < 2E\text{-}16$) internal rotation, 0.0070 mm ([-0.010,-0.004], $P = 8.07E\text{-}8$) medial translation, 0.021 mm ([-0.024, −0.018], $P < 2E\text{-}16$) posterior translation, and 0.0078 mm ([0.0063, 0.0093], $P < 2E\text{-}16$) superior translation difference between leg press and lunge (Fig. 5B). Across a 100° range of knee flexion, these results indicate the physical difference between these two tasks would be 0.58° in varus-valgus rotation, 2.4° internal-external rotation, 0.70 mm medial–lateral translation, 2.1 mm anterior-posterior translation, and 0.78 mm superior-inferior translation.

4. Discussion

We proposed a supine leg press as an alternative to the lunge that aims to provide a similar evaluation of knee motion, without the challenges of weightbearing deep-knee flexion, while increasing the measured range of motion. The use of the supine leg press was verified as an alternative to the standing lunge in a non-symptomatic adult cohort and differences between the tasks were identified.

Our kinematic results measured for the lunge activity were similar to prior in-vivo measurements made by other researchers (Galvin et al., 2018; Hamai et al., 2013; Leszko et al., 2011; Navacchia et al., 2017). Meta-analysis results of 164 non-symptomatic participants aged 25 to 61 supported our observation that the tibia rotates internally (mean difference = 4.6° [3.55°, 5.64°]) and becomes more valgus (1.9° to 3.3°) in deep-knee flexion (Galvin et al., 2018). Hamai et al. (2013) presented similar findings for 5 healthy young-adult males with 15° femoral external rotation relative to the tibia, tibial internal rotation, and a 5° valgus rotation across the entire range (85° to 150°) of weightbearing dynamic knee flexion. Similarly, Navacchia et al. (2017) had 7 healthy older adults perform a lunge and, on average, expressed increased valgus rotation (from 2° varus to 2° valgus), increased internal tibial rotation (from 10° to 15° internal), relatively static medial (~1 to 2 mm medial) and anterior (~2.5 mm anterior) translation, and increased inferior translation (from 1 to 5 mm inferior) with greater knee flexion. The results of these studies verify our observation of internal tibia rotation and increased valgus in deep-knee flexion. Small discrepancies likely result from differences in the sampled population, recorded range of motion, and the assignment of bone coordinate systems.

The differences between the supine leg press and the standing lunge were consistent between men and women with similar model trends and magnitudes (Figs. 4 & 5). The only meaningful functional difference between these two tasks is the 46.3° [-52.2, −40.4] greater range of motion in knee flexion during supine leg press (range of difference of lunge relative to leg press for varus-valgus: −1.1° [-1.9, −0.4]; internal-external: −6.2° [-8.1, −4.2]; medial–lateral: −1.5 mm [-2.0, −0.9]; anterior-posterior: −3.3 mm [-4.3, −2.3]; superior-inferior: −1.2 mm [-1.7, −0.7]; Table 2). Despite differences in magnitude, the flexion range of motion for lunge and leg press were significantly associated (Pearson's correlation = 0.2747 [0.0050, 0.5073]; $P = 0.0465$). This indicates that individuals with a larger range of motion during lunge were likely to have a larger range of motion during leg press.

The most notable difference between sexes was men being more varus than women during the leg press and lunge, which is supported by previous non-symptomatic, native knee research (Bellemans et al., 2012; Cooke et al., 1997; Deep et al., 2015; Hirschmann et al., 2019). Functionally, men (-0.03 ± 3.02°) had a 3.94° ([-5.63, −2.25], $P = 2.20E\text{-}05$) lesser peak-valgus angle (women: 3.91 ± 3.12°) and a 3.78° ([-5.67, −1.79], $P = 0.000329$) greater peak-varus angle than women (men: −6.01 ± 3.79°; women: −2.28 ± 3.22°) during leg press. Lunge results determined men (-0.96 ± 3.00°) to have a 3.62° ([-5.38, −1.85], $P = 0.000143$) lesser peak-valgus angle (women: 2.74 ± 3.27°) and a 3.70° ([-5.43, −1.97], $P = 7.77E\text{-}05$) greater peak-varus angle (men: −5.15 ± 3.00°; women: −1.53 ± 3.40°).

One limitation of the current study is that the studied cohort only included non-symptomatic, native-knee participants and did not include those with pathology or surgical repair. Future work is needed to assess kinematic differences and range-of-motion associations between the leg press and lunge in populations such as those with osteoarthritis and total-knee arthroplasty. It is important to note that our cohort consisted of middle-aged and older adults encompassing the age most associated with chronic knee pathology. Even so, the participant removed from analysis due to their knee pain from suspected osteoarthritis could only complete a 44.2° (max: 79.4°; min: 35.2°) flexion range of motion

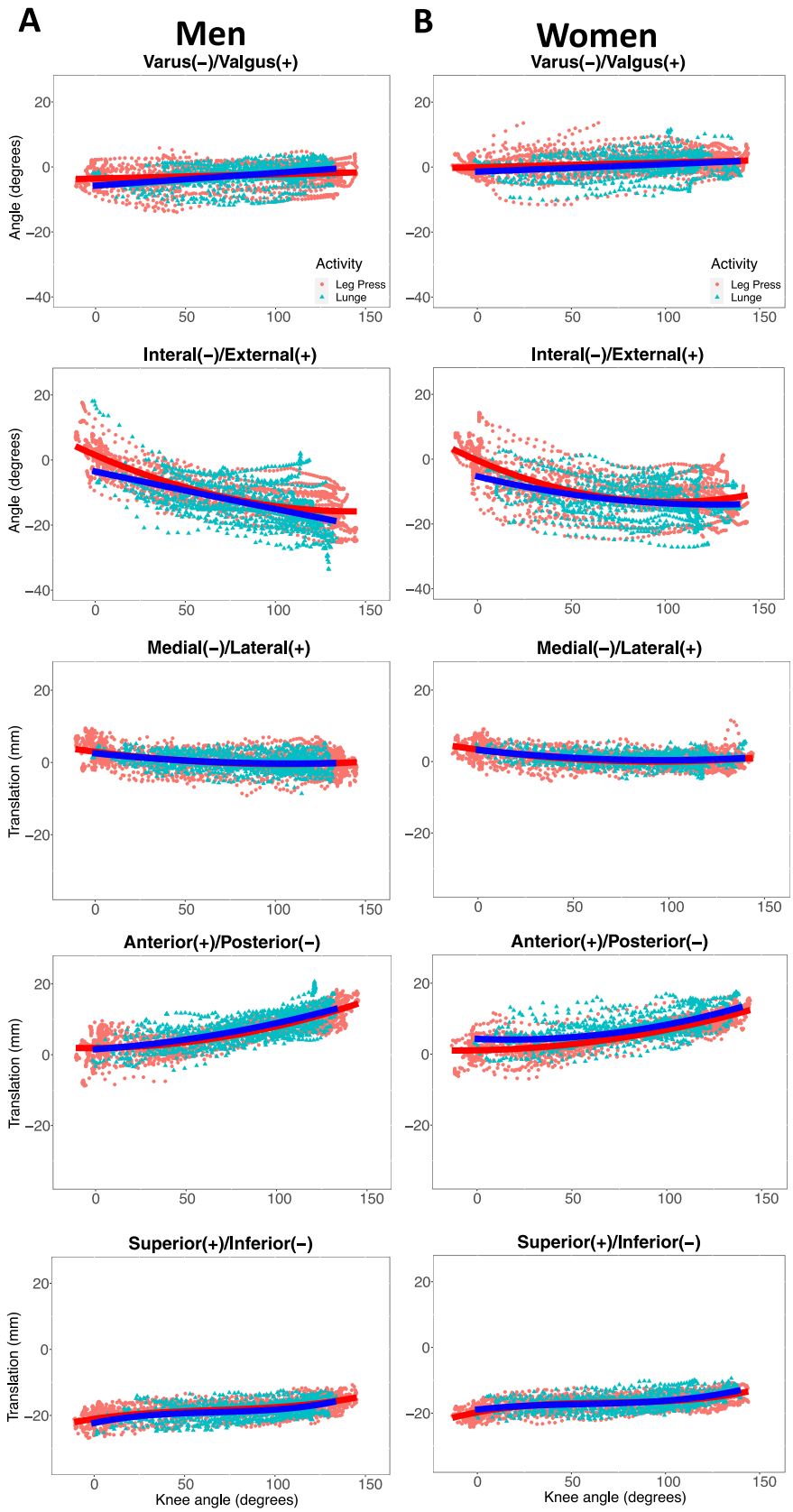

Fig. 4. Six degrees of freedom Grood and Suntay kinematics with regression fit: Red = Supine leg press; Blue = Standing lunge. (A) Kinematic results for men; (B) Kinematic results for women. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

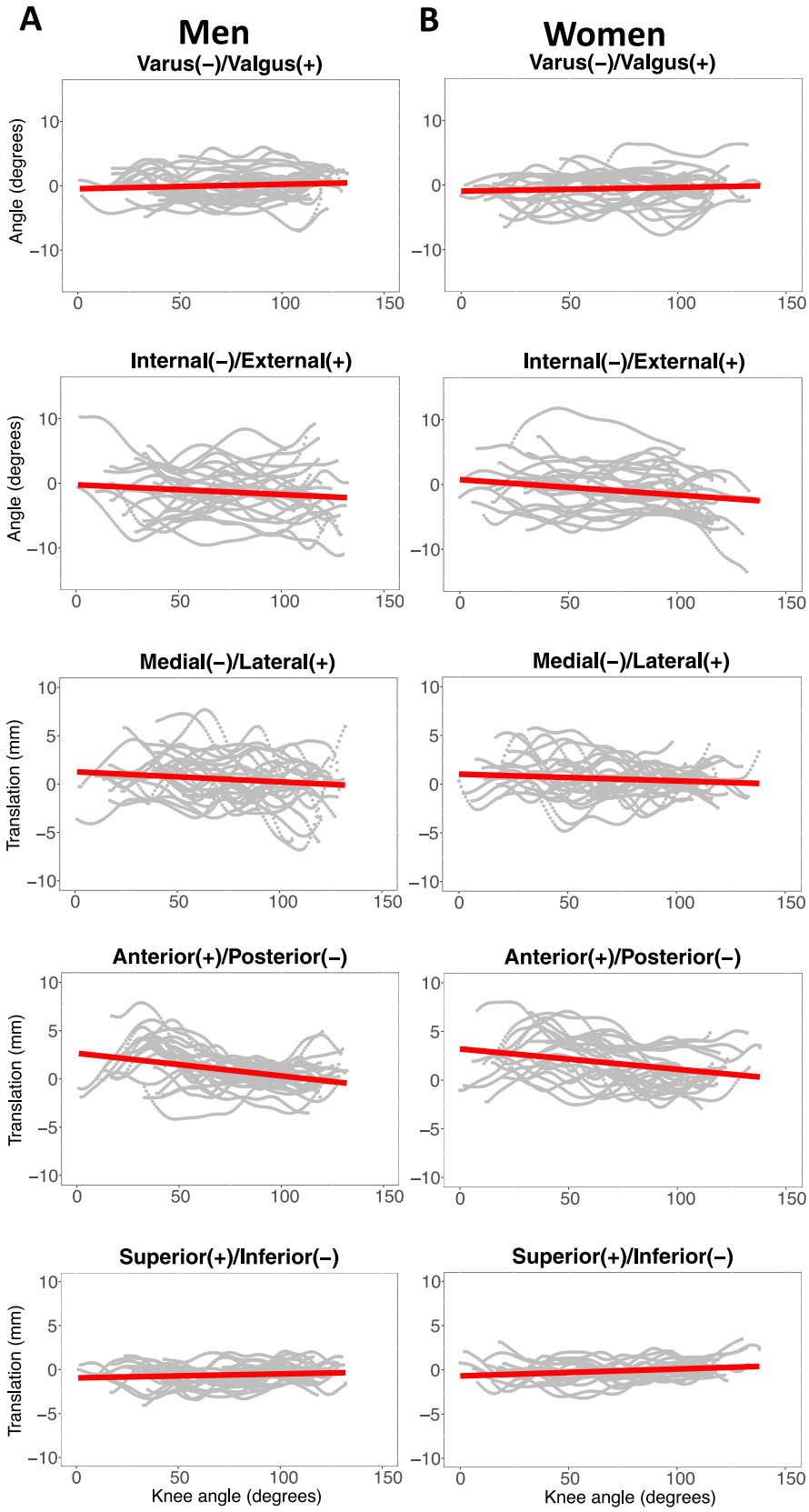

Fig. 5. Sex-specific models comparing kinematic differences between supine leg press and standing lunge. Gray = subject-specific kinematic difference; Red = linear regression fit. (A) Kinematic difference results for men; (B) Kinematic difference results for women. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

during the lunge but was able to complete a 123° (max: 128°; min: 5°) flexion range of motion during the leg press. This 78.8° difference highlights the potential for the leg press to be used by individuals with pathology, surgical repair, obesity, or other modalities that may limit the ability to successfully complete the lunge under their own control.

Despite the greater range-of-motion when performing the supine leg press, the task required two separate trials compared with one trial for the lunge. Performing two trials may increase the radiation exposure, though extremely low with pulsed radiography, and can potentially introduce discrepancies in kinematic profiles if participants are moved between trials. Future efforts plan to address this limitation by altering the HSSR system to allow the recording of the leg press in one trial.

Another limitation is the lack of repeated measures for each activity. Radiation exposure was calculated as part of a larger study that required the completion of six activities, including the leg press and lunge, and a lower-limb CT scan limiting our ability to repeat trials. However, all kinematic measures were treated as continuous and the entire range of motion was used for each participant providing a better analysis across the flexion range than quasi-static measurements. The larger sample size of 27 men and 26 women provided a sufficient representation to adequately apply the quantified and calculated differences in kinematics for each activity and sex, and robustly controlled for individual variability in kinematic performance. Future work will analyze multiple trials to address the reliability and repeatability of each task.

In conclusion, the supine leg press affords a worthwhile alternative to the standing lunge, while providing 46.2° greater range of motion in knee flexion, when trying to quantify knee kinematics. Notably, the measured increased knee flexion was present in a healthy cohort but was even greater for one subject with observed functional limitations due to pain and radiographic evidence of osteophytes. This result suggests potential application of the supine leg press for use in older adults and those with pathology and obesity. The similar kinematics, greater range of motion, and control of the applied load makes the supine leg press a suitable alternative for comparison of changes in knee motion across cohorts.

Data Statement

Due to the funding agreement of the research study, raw data was required to remain confidential and not shared. All regression results, coefficients [95% confidence intervals], and p values are provided as supplemental materials to offer greater context and transparency of the results.

CRediT authorship contribution statement

Landon D. Hamilton: Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Thor E. Andreassen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Casey Myers:** Writing – review & editing, Validation, Software, Methodology, Investigation, Funding acquisition, Conceptualization. **Kevin B. Shelburne:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Chadd Clary:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Paul J. Rullkoetter:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbiomech.2022.111118.

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